

- 1 (a)** As the rocket engine burns fuel, the exhaust gas is discharged from the back of the rocket and experiences a large change momentum towards the left. Since momentum of the whole system is conserved, the car will experience a similar magnitude of change in momentum towards the right. By Newton's second law, there is a forward force acting on the car towards the right. This force is produced due to the rocket engine burning the fuel.

- (b) (i) 1.** By drawing a tangent to the graph at $t = 2.0$ s, we have

$$\text{gradient of the graph} = \frac{12 - 4.4}{3.5 - 0.6} \approx 2.62$$

Since gradient of the graph gives the acceleration, $a = 2.62 \text{ ms}^{-2}$.

- 2.** By Newton's second law of motion